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Diffuser-incorporated transmission NIR measurement for reliable analysis of packed granular samples†

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A diffuser-incorporated transmission near-infrared (NIR) scheme that enables direct spectral collection of packed granular samples with reliable sample representation and reproducibility has been demonstrated. The analytical utility of this method has been evaluated for the determination of polyethylene (PE) pellet density and the discrimination of the geographical origin of rice samples. Based on the preliminary observation of transmission spectral features acquired from spherical polyoxymethylene (POM) packings composed of different particle sizes as well as packing thickness, a portion of the radiation was propagated through the void space in the packing without fully interacting with the POM pellets. This type of radiation, so-called non-fully interacted radiation (NFIR), adversely affected the sample representation as well as the reproducibility of transmission measurements. To maximize the interaction of NIR radiation with granular samples, a polytetrafluoroethylene (PTFE) diffuser was positioned in front of the sample packing to introduce isotropically diffused radiation into the sample. This diffuser-incorporated scheme resulted in highly reproducible transmission spectra for both packed granular samples. Consequently, the density determination of PE pellets as well as discrimination of rice samples according to geographical origin was more accurate using the proposed scheme.

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Introduction

To analyze solid samples such as powders, pellets, and granules using near-infrared (NIR) spectroscopy, spectral collection based on diffuse reflectance is normally employed since it enables simple and non-destructive analysis.[1–10] Solid samples are usually rotated or reciprocated during spectral collection to cover a larger sample volume, thereby achieving more representative sampling and reproducible measurement. However, diffuse reflectance spectra contain more weighted chemical information on constituents localized near the surface of samples, because the amount of NIR radiation entering a solid decreases exponentially as it propagates inside the sample.[11] This effect is especially problematic for the measurement of internally inhomogeneous samples. Nonetheless, diffuse reflectance measurement has been a common standard for NIR spectroscopic analysis of solid samples.

Transmission NIR measurement is a preferable technique for analysis of granular solid samples since acquired spectral features are fully representative of the entire sample. Several reports have compared the analytical performance of diffuse reflectance and transmission measurements for analysis of several solid samples such as pharmaceutical tablets[12–15] and fruits.[16,17] In most cases, transmission measurements provided a more accurate analysis, mainly due to improved sample representation. Despite its potential analytical superiority, transmission NIR measurements have been narrowly focused on the analysis of single solid samples, such as individual pharmaceutical tablets. The limited use of transmission measurement appears to be for two major reasons. First, the reliability of transmission NIR measurements through packed granule samples has not been thoroughly ensured for practical field applications. As observed with diffuse reflectance measurements, transmission spectral features can also be affected by diverse factors such as variations in sample packing and particle size. Since there is a considerable void space inside a granular packing, some illuminated radiation would partially interact with only the outer part of granules and/or propagate through the void space by reflecting from the granule surface without fully interacting with the sample. This type of radiation, so-called non-fully interacted radiation (NFIR), produces transmission spectra insufficient to represent the chemical identity of granule samples and impairs reproducibility of

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measurement. Thus, NFIR should be eliminated or at least minimized. Second, NIR radiation will be largely attenuated during propagation through a dense solid medium. This leads to substantial degradation of the signal-to-noise ratio of a collected spectrum, so possibly deteriorating measurement accuracy.

A novel diffuser-incorporated transmission NIR measurement that enables the collection of highly representative and reproducible spectra from packed granular samples is demonstrated in this study. It is designed to suppress NFIR by the generation of isotropically diffused NIR radiation using a diffuser (polytetrafluoroethylene (PTFE) disk) to maximize the interaction with granule samples in the packing. Initially, we characterized the degree of NFIR occurrence from transmission NIR spectra of spherical polyoxymethylene (POM) packings composed of different particle sizes as well as packing thicknesses. To collect transmission NIR spectra of packed POM, high power NIR radiation was illuminated onto a large sample area of 19.6 cm^2 (diameter: 5 cm). Variations of transmission spectral features were analyzed in relation to the percentage of void volume and packing thickness. The spectral features and reproducibility of POM spectra were also compared for spectral collection with and without the diffuser. In addition, the characteristics of light propagation inside the packing were also studied using Monte-Carlo simulation.

To evaluate the effectiveness of the diffuser-incorporated transmission scheme, we chose to analyze the density of polyethylene (PE) pellet samples and rice samples of different geographical origins. Transmission spectra were collected for these samples using the spectral collection scheme with and without the diffuser for comparison, and the spectral features in each case were closely examined to address diverse issues such as repeatability of spectral collection with random packing and spectral representation for packed samples. In addition, possible inhomogeneity within PE pellets and rice grains was investigated by examining microscopic NIR spectral features collected at different spots on the surface of sliced samples. Finally, determinations of PE pellet density and the discrimination of geographical origin of rice samples were performed using partial least squares (PLS) and principal component analysis-linear discriminant analysis (PCA-LDA), respectively, and the resulting accuracies of the measurements with and without the diffuser were then compared.

Experimental

PE pellets and rice samples

PE pellets of different densities employed in a previous publication[18] were used in this study. Three different sample grades from different polymerization processes were employed (density range: 0.918–0.961 g cm^{-3}). Fig. 1 shows top-view photographs of the packed PE pellets with 3 different densities (0.918 (a), 0.934 (b), and 0.961 g cm^{-3} (c)) in Petri dishes. The shape of the low-density pellets was disk-like, while, high-density pellets were more spherical. As shown, the packings have void spaces, with the high-density pellets being relatively more densely packed.

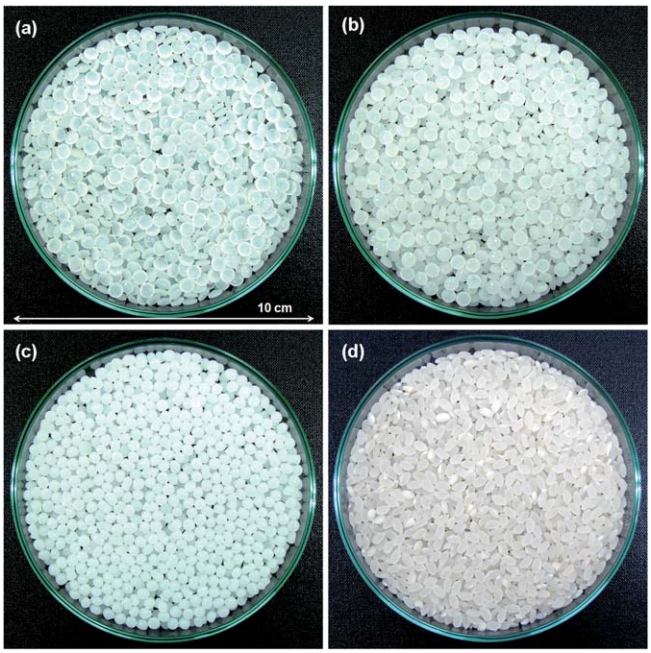

Fig. 1 Pictures (top-view) of packed PE pellets of 3 different densities (0.918 (a), 0.934 (b) and 0.961 g cm^{-3} (c)) and a packed Korean rice sample housed in a Petri dish (d).

Thirty Chinese and 30 Korean rice samples[19] collected from diverse regions in both countries were kindly supplied by the National Agricultural Products Quality Management Service (NAPQ), Seoul, Korea. The collection intended to include rice samples of different species cultivated in diverse regions in each country and continued over one year under careful supervision of the NAPQ. Fig. 1(d) shows a photograph of the packed Korean rice samples in a Petri dish. The rice packing was denser than the PE pellet packing.

Collection of NIR spectra

A high power (600 W) tungsten–halogen source (Sciencetech Inc., London, Ontario, Canada) was adopted to ensure sufficient transmission of NIR radiation through the sample packing. Radiation from the source was circularly collimated (diameter: 5 cm, area: 19.6 cm^2) to illuminate samples packed in a glass Petri dish. The wide area coverage allowed collection of transmission spectra to be more representative of the packed samples. The packing thickness for PE pellets and rice samples was approximately 7 mm. Roughly 450–550 PE pellets and rice grains were packed in the illumination volume. Finally, the transmitted radiation was gathered with a lens and delivered to an FT-NIR spectrometer (ABB Bomem) equipped with an InAs detector. Triplicate spectra were collected for each sample. In replicate spectral collection, samples were reloaded into the Petri dish.

Diffuse reflectance spectra of PE pellets and rice samples were also collected using an FT-NIR spectrometer from the same manufacturer. Samples were transferred into a glass vial (diameter: 2.5 cm) and diffuse reflectance spectra were collected while the sample vial was rotated and vertically reciprocated.

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Triplicate spectra were collected for each randomly packed sample. The resolution of both transmission and diffuse reflectance spectra was 8 cm^{-1}. All spectral pre-processing and multivariate analyses, including baseline correction, normalization, multiplicative scatter correction (MSC), PLS and PCA-LDA, were performed using MATLAB Version 7.0 (Math Works Inc., MA, USA). Monte-Carlo simulation was performed by Montcarl Version 2013, a free computational package.[20,21]

Results and discussion

Investigation of transmission NIR spectra for POM packings of different particle sizes

Initially, the variation of transmission spectral features of packed granules as a function of particle size and packing thickness was investigated using spherical polyoxymethylene (POM) particles as model granules. Spherical particles were chosen since the arrangement of individual particles in the packing was regular and the corresponding void volume could be simply calculated. The molecular structure of POM is shown in Fig. 2(a). POM particles of 3 different sizes (2.4, 3.2, and 4.0 mm in diameter) were packed in different thicknesses as described in Table 1, and NIR spectra were collected through these POM packings. The percent void volume calculated for each packing is also shown in the table.

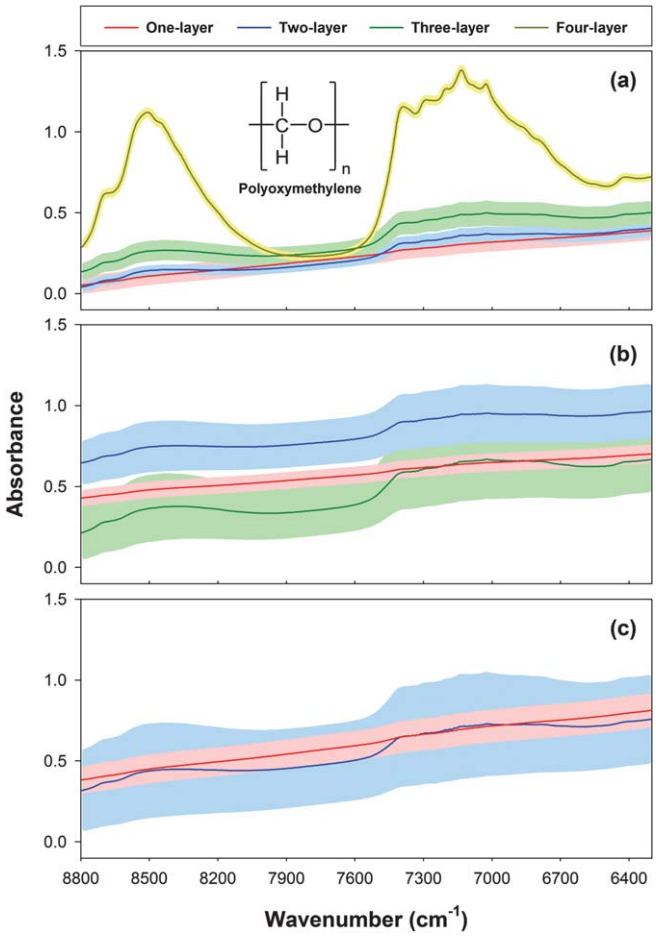

Fig. 2 Average transmission NIR spectra calculated from ten replicate measurements without the diffuser for POM packings of different particle sizes ((a): 2.4 mm, (b): 3.2 mm, (c): 4.0 mm in diameter). The shades indicate the corresponding standard deviations of ten replicate spectra.

Table 1 Percentage of void volumes in POM packings constructed with different particle sizes and thicknesses

Diameter (mm)	Number of packing layers	Packing thickness (mm)	Percentage of void volume (%)
2.4	1	2.4	48.6
2.4	2	4.4	44.9
2.4	3	6.5	43.5
2.4	4	8.6	42.8
3.2	1	3.2	46.8
3.2	2	5.9	43.0
3.2	3	8.7	41.6
4.0	1	4.0	48.8
4.0	2	7.4	45.1

In the packings of 2.4 mm particles up to 4 layers, the thickness varied from 2.4 to 8.6 mm and the void volume percentage decreased with the increase in packing layers (refer to Table 1). Fig. 2(a) shows NIR spectra of these packings collected without the diffuser. Average spectra obtained from ten replicate spectra collected with random repacking are shown for each case. The shades indicate the corresponding standard deviations of ten replicate spectra. For a one-layer packing, no noticeable absorption features were observed due to the direct passage of NIR radiation through the void space. For a two-layer packing, the second layer blocked some of the passing radiation through the void space, and weak broad absorption features started to appear. With the three-layer packing, the absorption magnitude further increased due to the increased packing thickness, while the spectral features were still not clear. This result implies that NFIR occurs even with the three-layer packing. Rather abruptly, the absorption features of POM became distinct with the four-layer packing, indicating an increased interaction of NIR radiation with the POM particles.

Fig. 2(b) shows average NIR spectra corresponding to ten replicate measurements for one-, two-, and three-layer packings of 3.2 mm POM particles. No significant spectral features were observed with the one-layer packing. The POM bands were still not apparent with the two- and three-layer packings, although the corresponding void volume percentages were similar or smaller than that of the four-layer packing of 2.4 mm particles. In the case of the packing with larger particle, the individual unit void space became larger; whereas, the number of unit void spaces in a given volume decreased. Therefore, the void volume percentage for packings of the two different particle sizes did not vary substantially. The increased unit void volume in the packing raised the probability of NFIR occurrence. This explains the indistinct spectral features even at the thickest packing with 3.2 mm particles. Fig. 2(c) shows average NIR spectra collected from one- and two-layer packings of 4.0 mm particles. In both of these cases, distinct POM bands were also absent, due to the presence of NFIR from the further increased unit void volume in the packings.

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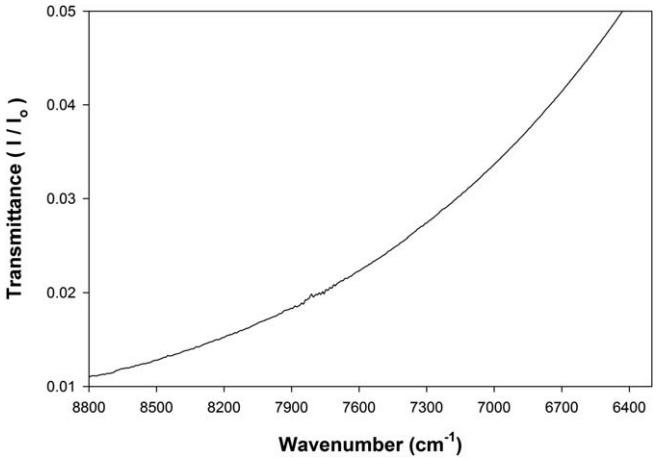

Fig. 3 Transmittance spectrum of a PTFE disk in the 8800–6300 cm^{-1} range. Air was used as a background.

Minimization or elimination of NFIR is necessary to collect reliable transmission spectra of packed granule samples. NFIR could be greatly diminished if isotropically diffused radiation such as that from a beam diffuser, rather than collimated radiation, illuminated the packing to maximize interactions of the radiation with packed granules. An optimal diffuser must generate sufficient isotropic radiation while having minimal absorption in the NIR range. To fulfill these requirements, a 1 mm-thick polytetrafluoroethylene (PTFE) disk was positioned in front of the Petri dish containing POM particles, and transmission spectra were recollected. As reported in several publications,[22,23] PTFE can diffuse a radiation in a nearly random-fashion and its optical properties are constant from ultraviolet (UV) to NIR region. So, the transmitted light through PTFE radiates like Lambert's cosine law, thereby providing isotropical diffusion. Fig. 3 shows the transmittance spectrum of a PTFE disk in the 8800–6300 cm^{-1} range. Air was used as a background. The transmittance increases exponentially as the wavelength of radiation becomes longer. This is due to the fact that the degree of scattering is a function of wavelength and diminishes for longer wavelength radiation. Other than the exponential-type transmittance pattern, no particular absorption feature is observed in the spectral range. The choppy feature around 7800 cm^{-1} originates from the absorption of gaseous water in air. As observed above, the PTFE disk greatly attenuates the intensity of radiation; therefore, the use of a high-intensity source is necessary to maintain a reasonable signal-to-noise ratio in final absorbance spectra. For this purpose, a 600 W-bulb was used as a NIR source in this study.

Fig. 4 shows average NIR spectra for the same POM packings collected with the diffuser. The shades indicate the corresponding standard deviations of ten replicate spectra. At first glance, the standard deviation of 10 replicate spectra collected for each of the 2.4 mm POM particle packings is much smaller (a), indicating superior reproducibility in the spectral collection. In addition, the absorption features of POM are clear with a good signal-to-noise ratio. Furthermore, absorbance clearly increased with packing thickness. The spectral features of the

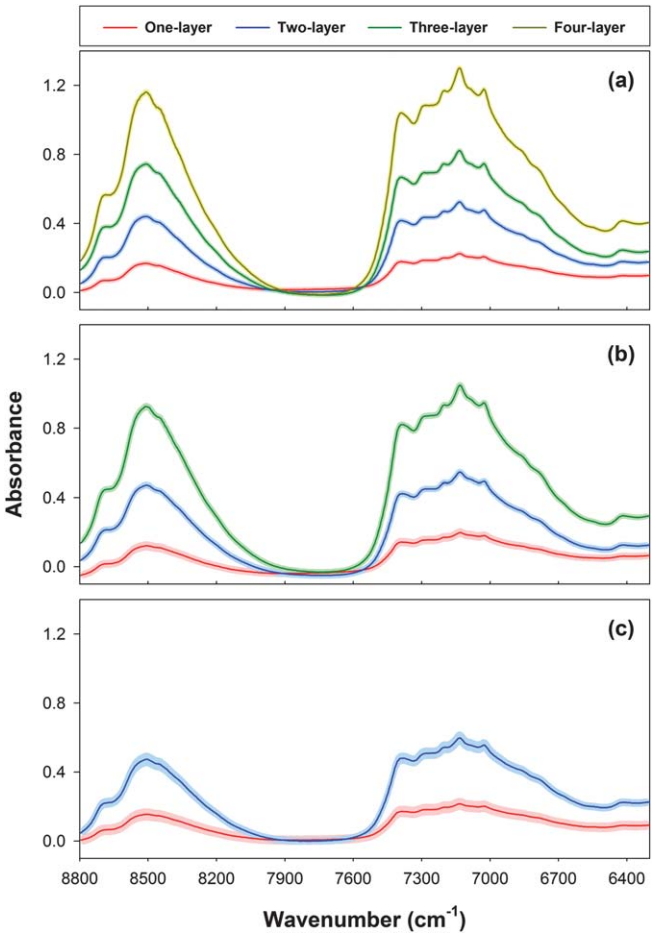

Fig. 4 Average transmission NIR spectra calculated from ten replicate measurements with the diffuser for POM packings of different particle sizes ((a): 2.4 mm, (b): 3.2 mm, (c) 4.0 mm in diameter). The shades indicate the corresponding standard deviations of ten replicate spectra.

3.2 and 4.0 mm POM particle packings were also distinct as shown in Fig. 4(b) and (c); whereas the slight baseline variation among the replicate spectra became noticeable in the spectra obtained from the packing of 4.0 mm particles. Since the unit void space in the 4.0 mm-particle packing was larger, the chance of NFIR occurrence could increase even using the diffuser, thus providing an explanation for the relatively larger baseline variation. Nonetheless, the POM spectral features were not degraded even with larger particle packings. Irradiation of isotropically diffused radiation to the packing led to the collection of highly representative and reproducible spectra for packed POM particles.

The absorption magnitude of the 4 layer-packing of 2.4 mm particles was greater than that of the 3 layer-packing of 3.2 mm particles, although both packing thicknesses were nearly the same as shown in Table 1. This result indicates that the interaction between NIR radiation and packed particles increased as the unit void space in the packing became smaller with smaller particles. This effect also explains the larger baseline variation and decreased absorption magnitude for spectra collected with the 2 layer-packing of 4.0 mm particles. Consequently, unit void

volume is a major factor governing the degree of NFIR occurrence. Hence, the use of a diffuser is indispensable for ensuring reproducibility in transmission spectral collection through packed granule samples.

Simulation of photon propagation inside POM packing based on the Monte-Carlo method

To investigate the propagation of NIR radiation through POM packing, a numerical simulation based on the Monte-Carlo method was performed. In the simulation, it was assumed that individual NIR photons propagated elastically and independently through the packing in a random walk-like fashion in a three-dimensional space, as similarly performed in modeling for fluorescence spectra of subsurface layers[24] and scattered photon inside bulk media in the measurement based on spatially offset Raman spectroscopy (SORS).[25] A simplified system is applied for the simulation here in that NIR photons propagate straight toward a designated sample and thereafter their direction is fully randomized upon the occurrence of scattering. Objects in the POM packing were separately defined by x, y and z in a Cartesian coordinate as shown in Fig. 5(a). POM particles were regularly packed with a layer thickness (d_2) and a PTFE disk with 1 mm thickness (d_1) was added in front of the packing. Refractive indices of POM, PTFE and the surrounding were 1.480, 1.350 and 1.000, respectively. The simulation considers that the sample is a semi-infinite medium with an air–medium interface located at $z = 0$.

The approximations made for the simulation were that (1) a wide area illumination (WAI) beam had a uniform intensity over 5 cm in diameter, (2) all incoming photons had equal probability of being injected at any point on the sample surface and (3) all photons coming out from the sample (at the sample-to-air interface) did not return to the sample. In addition, it was assumed that all NIR photons emerging from the last layer were detectable, although only a fraction of the photons is recognized within a solid angle acceptance of a detector in reality. The wavenumbers of injecting photons (one hundred thousand photons) varied from 8800 to 6300 cm^{-1} with an interval of 25 cm^{-1} and the number of detected photons was counted at each number. Finally, all photon count units were converted to absorbance units.

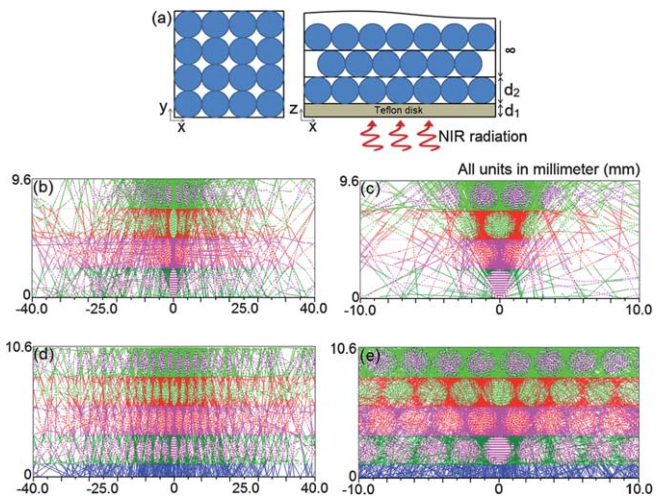

Fig. 5 Objects in POM packing separately defined by x, y and z in a Cartesian coordinate. POM particles are regularly packed with a layer thickness (d_2) and a PTFE disk with 1 mm thickness (d_1) is located in front of the packing (a). Simulations of light propagation (8510 cm^{-1}) in the xz plane of a 4-layer packing composed of 2.4 mm POM particles without (b) and with (d) use of the diffuser are shown. The bottom blue layer corresponds to the diffuser. Both simulation features are magnified in the x-span of 20 mm (without (c) and with (e) the diffuser).

Fig. 5(b) shows the simulation of light propagation (8510 cm^{-1}) in the xz plane of a 4-layer packing composed of 2.4 mm POM particles when incident radiation hits the packing without passing through the diffuser. The span of the x-direction is 80 mm (−40 to 40 mm). Each layer was designated using different colors for easier recognition. All dimensions were in millimeters (mm) and the diameter of NIR illumination was 50 mm. The lines indicate the propagation of radiation and the darker color in a particle implies more interactions with the radiation. Fig. 5(c) presents the magnified simulation result in the x-span of 20 mm (−10 to 10 mm). It is important to note that the light propagation shown here corresponds to just one snapshot out of tremendous possibilities. As shown, the particles located at the center and back of the packing interact much more with the radiation. Since the direction of scattering in a particle is random, the scattered radiation has a relatively higher chance of being concentrated along the center line of the packing when collimated radiation irradiates the packing directly without use of the diffuser.

Fig. 5(d) shows the simulation of light propagation (8510 cm^{-1}) in the xz plane of a 4-layer packing composed of 2.4 mm POM particles when incident radiation hits the packing through the diffuser and the magnified feature is shown in Fig. 5(e). The bottom blue layer corresponds to the diffuser. The pattern of light propagation is substantially different when the diffuser is used. As shown, the radiation interacts with large numbers of POM particles. The irradiation of diffused radiation significantly increases the chance of photons to interact with more particles across the packing; whereas those in the x-span from −10 to 10 mm are relatively more interacted.

Absorbances at each wavenumber were calculated for single as well as multilayer packings composed of 2.4, 3.2 and 4.0 POM particles. Out of all calculations, Fig. 6 shows the experimental (a) and calculated absorbance (b) at 8510 cm^{-1} for 1- to 4-layer packings of 2.4 mm POM particles in the case of spectral collections without (blue) and with (red) the diffuser. At each packing, the calculated absorbance in the absence of the diffuser is always lower compared to that with the diffuser in place due to the presence of NFIR as expected. Although absolute magnitudes of absorbance at each packing are slightly different in both cases, the variation of calculated absorbance according to the increase of the packing layer generally resembles the experimental observation. In the measurements of 2- and 3-layer packings without a diffuser, the calculated absorbances are fairly higher compared with the corresponding experimental absorbances. The POM surface is polished and so moderately shiny. The shiny surface increases the occurrence of

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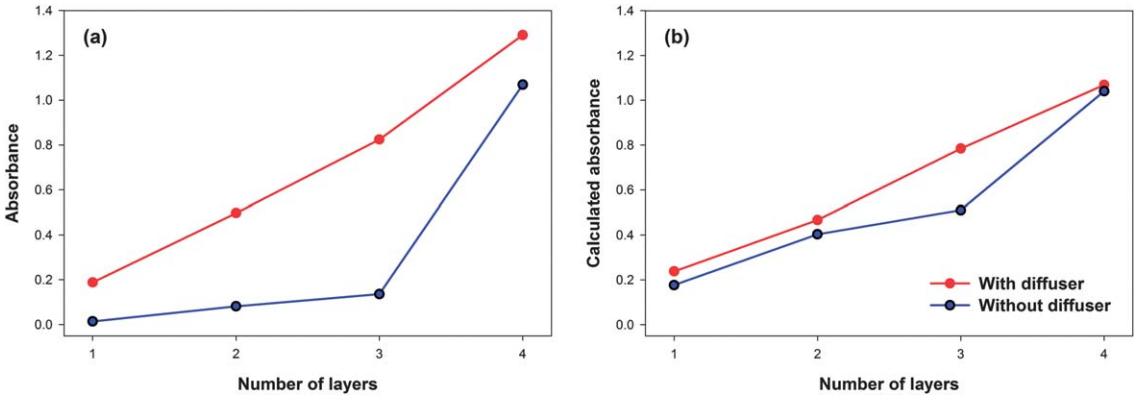

Fig. 6 Experimental (a) and calculated absorbances (b) at 8510 cm^{-1} for 1- to 4-layer packings of 2.4 mm POM particles in the case of spectral collections without (blue) and with (red) the diffuser.

NFIR, thereby decreasing the resulting absorbance. In the simulation, the reflection occurring at the polished surface is not considered; therefore, the calculated absorbances are higher.

Furthermore, to explain the improved reproducibility of the diffuser-incorporated measurement, the simulation was repeated 10 times and the standard deviations of calculated absorbances at 8510 cm^{-1} for both cases were compared as shown in Fig. 7. As expected, in all cases, the standard deviations are smaller for the spectral collection with the diffuser in place. Overall, the simulation results based on the Monte-Carlo method clearly demonstrate the substantial increase of photon interaction with POM particles by using the diffuser and the improved reproducibility for NIR spectral collection through POM packings.

Density determination of PE pellets using transmission measurement

Fig. 8 shows raw (a) and normalized (b) NIR spectra of a high-density sample (density: 0.961 g cm^{-3}) collected without the diffuser. Six replicate spectra collected with random repacking are shown. To effectively compensate for the variation in absorbance caused by different effective pathlengths for each spectral collection, initially, the baseline was linearly corrected at 8800, 7480 and 6300 cm^{-1}, and then further normalized by dividing by the corresponding integrated area calculated in the 8800–6300 cm^{-1} range. The peaks in the 8800–7900 and 7400–6400 cm^{-1} ranges corresponded to the second overtone and combination bands, respectively, of mostly CH_2 (methylene) and CH_3 (methyl) groups.[26,27]

As shown in Fig. 8(a), the baselines of spectra varied vertically. When NFIR occurred, the amount of radiation passing through the pellet packing varied with the random repacking, which induced the observed vertical baseline variation. The bands around 8220 cm^{-1} in the bottom two raw spectra were less distinct than in other spectra, demonstrating unsuccessful sample representation caused by NFIR. Even after normalization (Fig. 8(b)), the absorption magnitude varied, and spectral features, especially around the 8220 cm^{-1} band, were slightly different.

Fig. 8(c) and (d) show raw and normalized NIR spectra (six replicates) of the same high density pellets collected with the diffuser in place. The considerable baseline variation observed in Fig. 8(a) nearly disappeared. Spectral features were distinct and very reproducible after normalization. For the closer examination, the bands around 8220 cm^{-1} were magnified as shown in the inset. These results indicate the successful

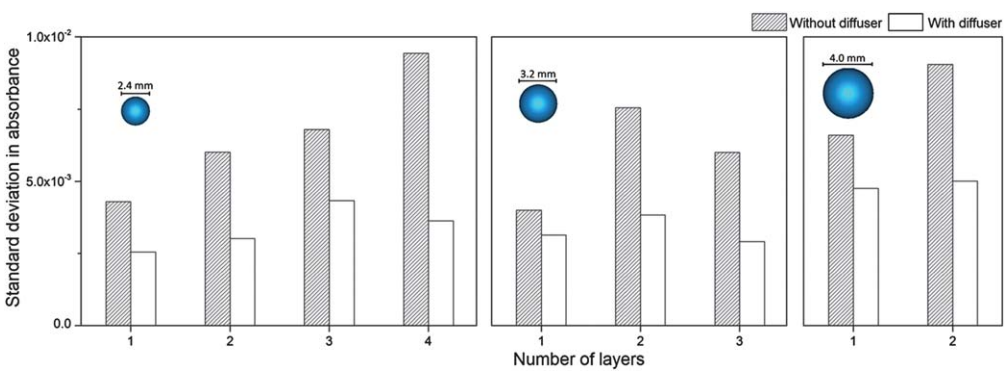

Fig. 7 Standard deviations of calculated absorbances at 8510 cm^{-1} when the simulation is repeated 10 times for spectral collection of each POM packing without and with the diffuser.

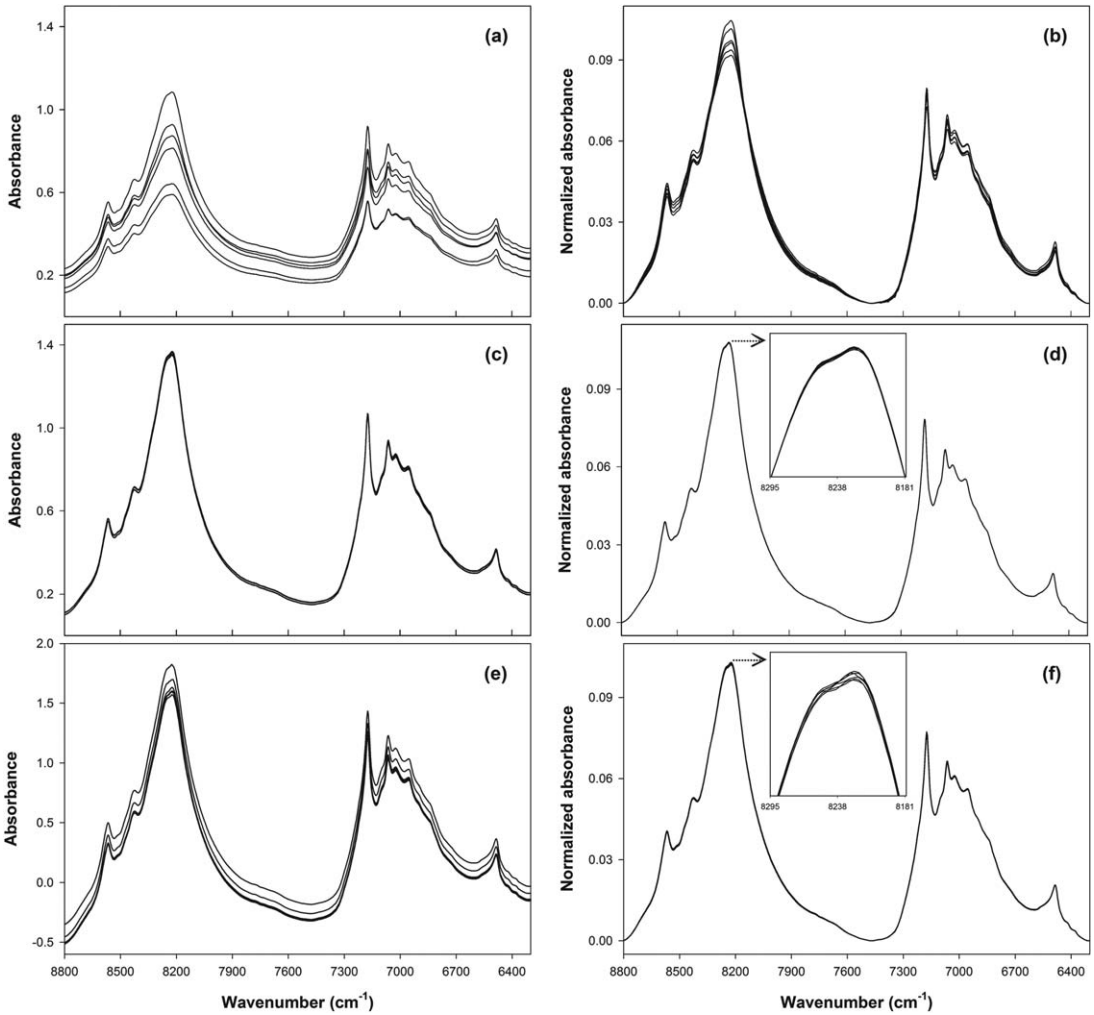

Fig. 8 Raw (a) and normalized (b) NIR spectra of PE pellets (density: 0.961 g cm^{-3}) collected without the diffuser. Raw (c) and normalized (d) NIR spectra of the same sample collected with the diffuser. Raw (e) and normalized (f) NIR spectra of PTFE-filled PE pellets collected without the diffuser. In all cases, six replicate spectra are shown. Both insets magnify the bands around 8220 cm^{-1}.

suppression of NFIR using the PTFE diffuser. Since the reference density of PE samples was known, the volume occupied by PE pellets was calculated using the measured mass of the packing. The void volume percentages were calculated for three different PE pellet packings as shown in Fig. 1. The calculation was repeated 10 times for 10 random packings. The void volume percentages for the low, middle, and high-density PE samples were 26.3 ± 0.2, 24.5 ± 0.5, and 21.8 ± 0.2%, respectively.

Filling of a PTFE powder in the void space is an alternative way to minimize the occurrence of NFIR and so it is worthy of investigation. For this purpose, the void space in the packing of the high-density sample (0.961 g cm^{-3}) was filled with a 35 μm-PTFE powder and the corresponding transmission spectra were collected without use of the diffuser. The procedure for the repacking of the pellets and filling of the PTFE powder was repeated 6 times. Fig. 8(e) and (f) show raw and normalized NIR spectra (six replicates) of the PTFE-filled high-density pellets, respectively. The inset in Fig. 8(f) also highlights the bands around 8220 cm^{-1}. Two noticeable observations in the raw spectra are that absorbance is larger and the vertical baseline variation is relatively greater in comparison with the spectra in Fig. 8(c). The increased absorption is from the scattering induced by the filled powders, thereby increasing the chance of NIR radiation to interact with individual pellets and resulting in the increased optical pathlength. The vertical baseline variation implies that the filling of the PTFE powder in the void space may not be consistent when it is repeated. Nevertheless, the obtained spectra are quite reproducible; so the spectral features are similar to each other after normalization (Fig. 8(f)).

The PTFE powder-filling in the void space has several disadvantages based on a viewpoint of the practical spectroscopic analysis. The filling procedure and recovery of the powder by sieving substantially lengthen analysis time, and so the overall practice is inefficient. Moreover, the recovery of the PTFE powder after sieving was 99.2% and this non-100% recovery was from the adsorption of the powder on the surface of PE pellets. Consequently, rather than the powder-filling, the placement of the diffuser in front of packed PE pellets was simple without requiring further sample pretreatment steps and effective to collect reproducible spectra. Therefore, the

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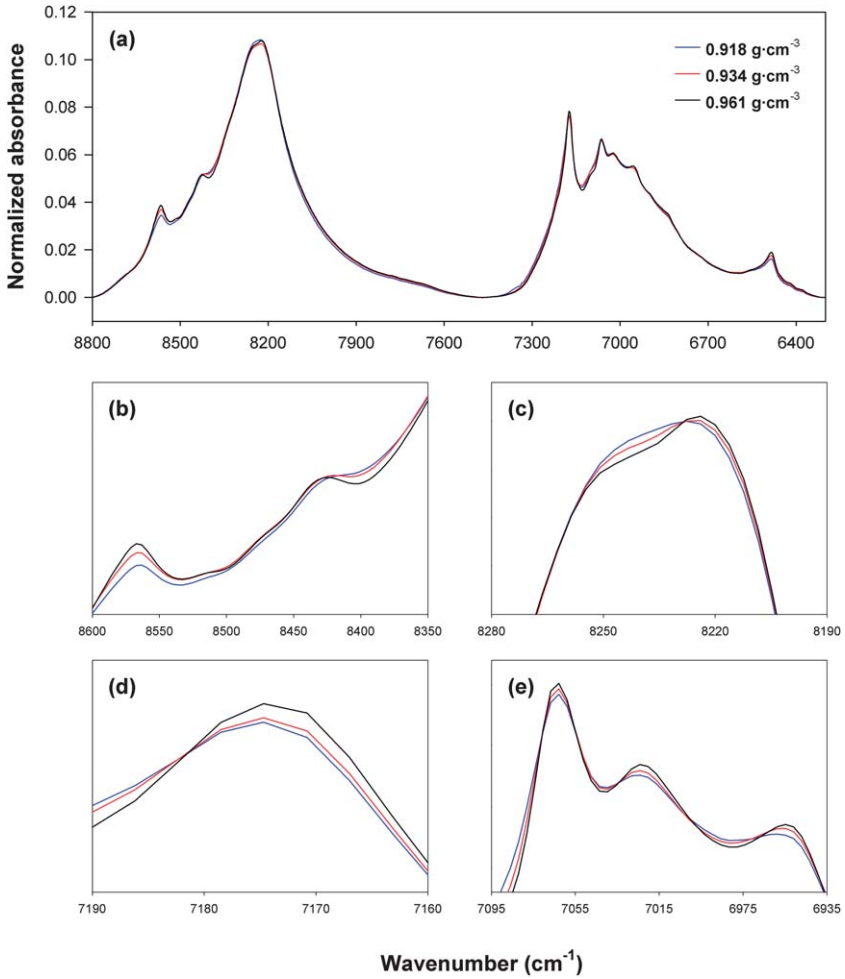

Fig. 9 Normalized transmission spectra of three PE samples with different densities (0.918, 0.934, and 0.961 g cm^{-3}) in the 8800–6300 (a), 8600–8350 (b), 8280–8190 (c), 7190–7160 (d) and 7095–6935 cm^{-1} (e) ranges.

diffuser-incorporated transmission scheme was solely employed for further NIR spectroscopic analysis.

Normalized transmission spectra (8800–6300 cm^{-1} range) corresponding to PE samples (density: 0.918, 0.934, and 0.961 g cm^{-3}) collected with the diffuser are shown in Fig. 9(a). The spectral features in four narrow spectral ranges, 8600–8350 (b), 8280–8190 (c), 7190–7160 (d), and 7095–6935 cm^{-1} (e), are also shown to closely examine the spectral changes according to the variation in PE density. Each PE spectrum in the figure corresponds to an average of 6 replicate spectra.

For the high-density PE, polymer molecules were arranged more orderly in a given volume (more crystalline). Therefore, the bands were relatively sharper with higher absorption, as shown in Fig. 9(b), (d), and (e). The same absorption bands for two lower density samples were broader with lower absorption. The less ordered (more amorphous) structures of the lower density samples led to broader spectral features.

The variation of absorption bands in the 8280–8190 cm^{-1} range differed from those in the other ranges (refer to Fig. 9(c)). The relative intensities of the 8248 and 8220 cm^{-1} bands, corresponding to the second overtone band of CH_3 and CH_2,[26,27] respectively, changed with the density variation. Increased CH_3 in the pellet corresponded to a more branched PE structure and a less compact polymeric arrangement, thus leading to a lower PE density. Therefore, the 8248 cm^{-1} band was more dominant for the low-density sample (0.918 g cm^{-3}). With an increase in density, the CH_2 band at 8220 cm^{-1} became more prominent, while the CH_3 band at 8248 cm^{-1} became weaker. Overall, the transmission NIR spectral features clearly reflect the variation of PE density.

PLS was used to determine the PE pellet density using both transmission NIR spectra collected with and without the diffuser. The 8800–6300 cm^{-1} range was used for PLS. As performed in a previous study, the resulting errors were obtained by fully cross-validating all samples in each dataset (20 samples assigned to calibration and 5 samples assigned to validation sets, respectively, for all possible combinations).[18] The resulting mean standard error of calibration (MSEC) and the mean standard error of prediction (MSEP) for measurement without the diffuser were 0.0019 and 0.0027 g cm^{-3}, respectively, while they were 0.0007 and 0.0012 g cm^{-3} for measurement with the diffuser. Six factors were used for both cases. The MSEP was dramatically improved using the diffuser-incorporated spectral collection scheme, since the obtained transmission spectra

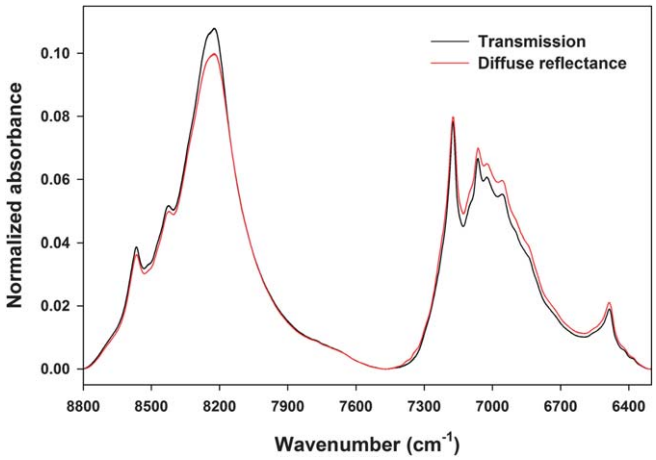

Fig. 10 Normalized NIR spectra of the high-density PE pellets (0.961 g cm^{-3}) collected by the diffuser-incorporated transmission (black) and diffuse reflectance (red) schemes.

were highly reproducible with clear representation of packed PE pellets.

Additionally, for the purpose of comparison, diffuse reflectance NIR spectra were collected for the same PE pellets and used to determine the densities, as performed by the Ozaki group who exploited NIR diffuse reflectance measurement of PE pellets for the prediction of their physical properties including density.[28] Fig. 10 shows normalized NIR spectra of the high-density PE pellets (0.961 g cm^{-3}) collected using the diffuser-incorporated transmission (black) and diffuse reflectance schemes (red). The absorption of the transmission bands in the 8800–7500 and 7500–6300 cm^{-1} ranges is relatively higher and lower, respectively, than those acquired from the diffuse reflectance measurement. The penetration depth of radiation into a sample in diffuse reflectance measurements varies as a function of wavelength, such as the deeper penetration for the longer wavelengths.[11] Therefore, in the diffuse reflectance spectrum, the absorption in the 8800–7500 cm^{-1} (shorter wavelength region) and 7500–6300 cm^{-1} ranges (longer wavelength region) is smaller and larger, respectively, compared to that in the transmission spectrum.

To determine the densities, multiplicative scatter correction (MSC) was applied to the diffuse reflectance spectra and then PLS was performed using the same 8800–6300 cm^{-1} range. The resulting MSEP was 0.0015 g cm^{-3} using 6 factors, which was greater than the MSEP using the diffuser-incorporated transmission measurement. The improved accuracy using the diffuser-incorporated transmission measurement was due to the better spectral representation of PE samples, which had the property inhomogeneity in a pellet as found in a previous publication.[29] Since there is a gradual structural variation from the surface to the center of each pellet, transmission spectral collection capable of fully representing the whole pellet is crucial for accurate analysis. In contrast, diffuse reflectance spectra of PE pellets could contain more weighted information relevant to the outer part of the pellet, causing minute discrepancies between the acquired spectral information and the corresponding PE properties. This discrepancy provides an explanation for the inferior accuracy in the PE pellet density determination when the diffuse reflectance scheme was used for spectral collection. The micro-inhomogeneity was also confirmed by collecting NIR spectra across a sliced pellet using a microscope and examining the corresponding spectral variation. The result is summarized in the supplementary data (refer to Fig. S-1†).

Determination of the geographical origin of rice samples

To further demonstrate the analytical potential of the diffuser-incorporated transmission scheme, discrimination of rice samples from two different geographical origins (Korea and China) was attempted. Initially, the void volume percentage inside rice packings was calculated. The average density of rice grains, calculated by measuring the increased volume after the addition of a known mass of grains into water in a graduated cylinder, was determined to be 1.43 g mL^{-1}. From the calculated density, the void volume percentage was calculated to be 14.6 ± 1.1% based on the measurement of six different rice sample packings.

Fig. 11 shows triplicate transmission NIR spectra (10 600–7700 cm^{-1}) collected for Korean (black) and Chinese (red) rice samples without (a) and with (b) the diffuser. Baselines of spectra collected without the diffuser varied greatly. In contrast, the magnitude of baseline variation was reduced by using the diffuser. For normalization, the baselines of spectra were corrected at 10 600 and 8900 cm^{-1}, and the baseline-corrected spectra were divided by the corresponding integrated areas calculated in the 10 600–7700 cm^{-1} range. Fig. 11(c) and (d) show normalized spectra corresponding to the raw spectra shown in Fig. 11(a) and (b), respectively. After normalization, the red-colored spectra were still not reproducible (Fig. 11(c)), indicating the occurrence of NFIR in the rice packing even though it was more densely packed. However, with the diffuser, normalized triplicate spectra corresponding to each geographical origin were reproducible.

As shown in Fig. 11(d), spectral differences between the Korean and Chinese rice samples were insignificant. The NIR features in the figure were primarily from starch, since it is the major component (80–84 wt%) of rice grains. For comparison, the NIR spectrum of starch (blue color, purchased at a local market) is also shown in the figure. Since starch is a long chain composed of repeating glucose units, its NIR spectral features would be analogous regardless of geographical or botanical origin. Alternatively, other rice constituents such as protein or fat would provide relatively more selective information for discrimination due to the larger structural variability. However, their contents are low (5–7 wt%). Therefore, NIR spectral differences between rice samples of the two origins were expected to be small as observed in Fig. 11(d).

Using the spectral datasets collected without and with the diffuser, discrimination between Korean and Chinese rice samples was performed using LDA. Initially, PCA was performed by combining the spectra from the rice samples of both origins, and the resulting scores were used for LDA. The first through sixth scores were used and an optimal combination of

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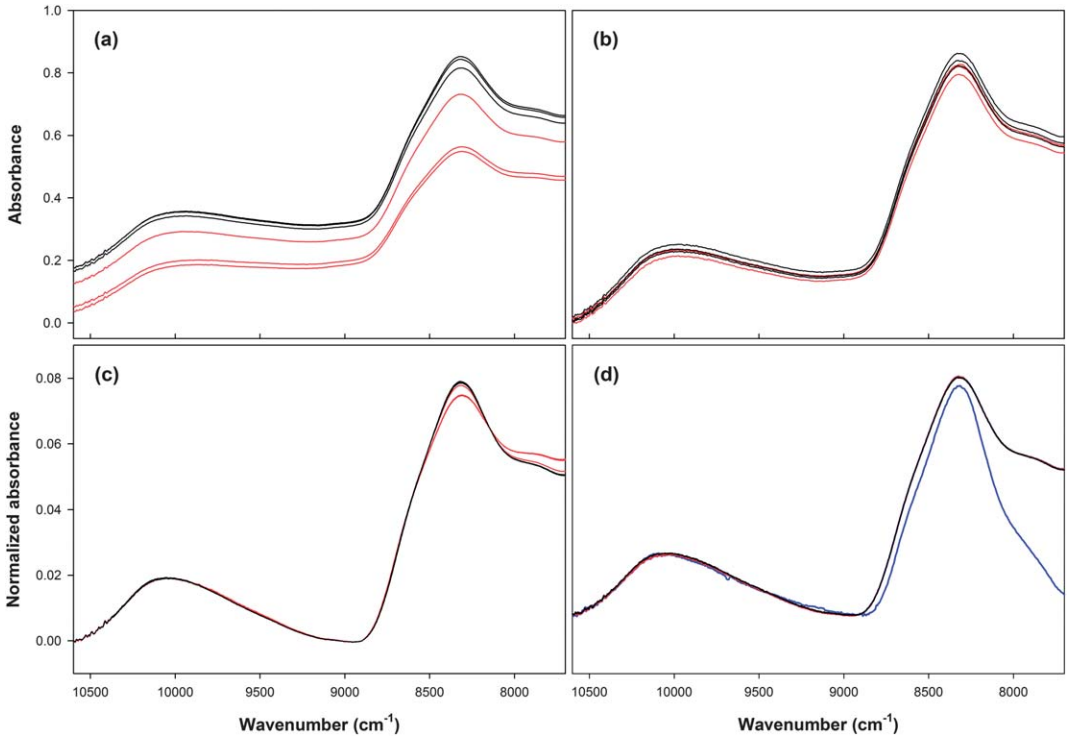

Fig. 11 Transmission NIR spectra (10 600–7700 cm^{-1} range) collected for one Korean (black) and Chinese (red) rice samples without (a) and with (b) the diffuser. The corresponding normalized spectra are shown in (c) and (d), respectively. For comparison, the NIR spectrum of starch (blue color) is also shown (d).

them yielding the best discrimination accuracy was searched. Discrimination errors were obtained by a leave-one-out cross-validation method. The resulting lowest errors for the measurement without and with the diffuser were 22.2 and 8.4%, respectively. In the case of diffuser-absent measurement, the combination of second/third/sixth scores for LDA provided the best accuracy; whereas the employment of second/third/fourth scores led to the minimal discrimination error in the diffuser-incorporated measurement. The corresponding three-dimensional score scatter plots yielding the best discrimination accuracy for the measurements without and with the diffuser are shown in the ESI (Fig. S-2†). The reproducible spectral features of packed rice samples achieved by the diffuser-incorporated measurement enabled minute spectral differences to be recognized, which improved the sample discrimination. The average spectra of both geographical origins and corresponding standard deviations are provided in the supplementary data (refer to Fig. S-3†).

For comparison, diffuse reflectance NIR spectra were also collected for the samples, and PCA-LDA was performed to discriminate the geographical origins using the MSC-treated spectra in the 10 600–7700 cm^{-1} range. The lowest discrimination error was 12.7% when the second/third/fourth scores were combined. Again, the diffuser-incorporated transmission measurement provided more accurate geographical discrimination compared to the diffuse reflectance measurements. As is known, the rice constituents such as embryo and endosperm are not homogeneously distributed in a grain. Therefore, acquisition of NIR spectra fully representing all of the constituents in individual grains is essential for accurate analysis. Thus, internally localized constituents in the grain such as the embryo would be relatively less represented in the diffuse reflectance spectra. In contrast, transmission spectra equally representing all of the grain constituents are more descriptive for the discrimination of compositional differences between the samples from two geographical origins. The heterogeneous distribution of constituents was confirmed by NIR microscopic investigation on a sectioned rice grain. The detailed result is further described in the ESI (refer to Fig. S-4†).

Conclusions

The diffuser played a crucial role in producing reproducible and representative transmission NIR spectra for PE pellet and rice grain packings. Sufficient interaction of isotropically diffused radiation with the packed samples effectively suppressed NFIR, leading to more reliable transmission NIR analysis through the packing of these granular samples. Moreover, the determination of density and discrimination of geographical origin using transmission measurements were more accurate compared to diffuse reflectance measurements due to improved compositional representation of individual granular samples. The successful introduction of the diffuser-incorporated scheme in this study could catalyze the use of transmission NIR analysis for diverse granular as well as powder samples in the future.

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